Today

- One-dimensional systems
- The infinite square well

2 One-dimensional systems

Now we begin constructing the wavefunctons for a system containing a single particle of mass m in one dimension. Different systems correspond to different potentials. We will be considering only potentials that are time-independent. The wavefunctions and the corresponding energies are then obtained by solving the time-independent Schrödinger equation in one dimension,

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x). \tag{2.0.1}$$

Once we find the wavefunction $\psi_E(x)$ of energy E, its time dependence follows,

$$\psi_E(x,t) = \psi_E(x)e^{-\frac{i}{\hbar}Et}. \tag{2.0.2}$$

2.1 Boundary conditions

The boundary conditions we impose depend on the potential. Suppose we are given a generic potential $V(x)$. Consider a point $x = a$ on the $x-$axis. We want to investigate whether the wavefunction $\psi(x)$ and its first derivative $\psi'(x)$ are continuous or not across $x = a$. First, rewriting the Schrödnger equation as

$$\psi''(x) = \frac{2m}{\hbar^2}(V(x) - E)\psi(x), \tag{2.1.1}$$

we note that the continuity or discontinuity of $\psi''(x)$ follows the continuity or discontinuity of the potential $V(x)$.

For the first derivative, and considering a particle of finite energy E,

$$\begin{aligned}\lim_{\epsilon\to 0}(\psi'(a+\epsilon) - \psi'(a-\epsilon)) &= \lim_{\epsilon\to 0}\int_{a-\epsilon}^{a+\epsilon}\psi''(x)dx = \lim_{\epsilon\to 0}\int_{a-\epsilon}^{a+\epsilon}\frac{2m}{\hbar^2}(V(x)-E)\psi(x)dx \\ &= \lim_{\epsilon\to 0}\frac{2m}{\hbar^2}\int_{a-\epsilon}^{a+\epsilon}V(x)\psi(x)dx,\end{aligned} \tag{2.1.2}$$

Now if $V(x)$ has a finite jump at $x = a$, this integral vanishes and $\psi'(x)$ is continuous across $x = a$.

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On the other hand, if $V(x)$ has an infinite jump, similar to the delta function, the integral will be nonzero but finite, and $\psi'(x)$ is discontinuous across $x = a$.

For the wavefunction itself,

$$\lim_{\epsilon\to 0}(\psi(a+\epsilon)-\psi(a-\epsilon)) \quad = \quad \lim_{\epsilon\to 0}\int_{a-\epsilon}^{a+\epsilon}\psi'(x)dx. \tag{2.1.3}$$

Now whether $V(x)$ has a finite or infinite jump at $x = a$, this integral vanishes, since $\psi'(x)$ could have only a finite jump across $x = a$. Therefore, $\psi(x)$ is always continuous across $x = a$.

We summarize the boundary conditions as follows

1. When the potential $V(x)$ has a finite jump at $x = a$, both $\psi(x)$ and $\psi'(x)$ are continuous across $x = a$.
2. When the potential $V(x)$ has an infinite jump at $x = a$, $\psi(x)$ is continuous but $\psi'(x)$ is discontinuous across $x = a$.

Furthermore, the wavefunction must vanish at $x = \pm\infty$ for a normalizable wavefunction.

2.2 Infinite square well

Consider a particle of mass m in the potential

$$V(x) = \begin{cases} 0, & 0 \le x \le a \\ \infty, & x < 0 \text{ and } x > a. \end{cases}$$

The potential is shown in the figure below.

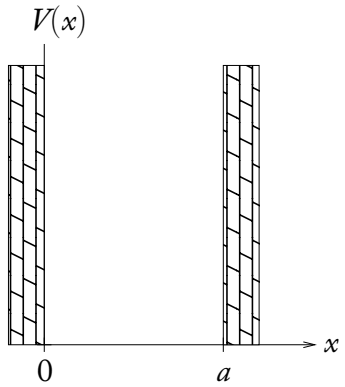


Figure 2.2.1: Infinite square well potential.

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Because, the particle cannot have infinite energy, it cannot be found in the regions $x < 0$ and $x > a$. Therefore,

$$\psi(x) = 0, \quad \text{for } x < 0 \text{and } x > a. \tag{2.2.1}$$

For the region in the well, $0 \leq x \leq a$, we have $V(x) = 0$ and the Schrödinger equation reduces to

$$\psi''(x) + k^2\psi(x) = 0, \tag{2.2.2}$$

where

$$k = \sqrt{\frac{2mE}{\hbar^2}} \tag{2.2.3}$$

and $E > 0$. We can write the general solution in terms of linear combinations of real solutions $\sin kx$ and $\cos kx$ as

$$\psi(x) = A \sin kx + B \cos kx. \tag{2.2.4}$$

Applying the boundary condition on the continuity of $\psi(x)$ at $x = 0$ sets $\psi(0) = B = 0$. We then have $\psi(x) = A \sin kx$, and applying the boundary condition at $x = a$ sets $\psi(a) = A \sin ka = 0$. Therefore, either $A = 0$ or $ka = n\pi$, $n = 1, 2, ...$ But $A = 0$ sets $\psi(x) = 0$ everywhere, which corresponds to no particle anywhere; i.e., $\psi(x, t) = 0$ is a solution to the Schrödinger equation for an infinite square well with no particle. Therefore,

$$k_n = \frac{n\pi}{a}, \quad n = 1, 2, ... \tag{2.2.5}$$

Question 2.2.1: Why are we not including zero and negative integers for n?

Answer: Because $n = 0$ sets $\psi(x) = 0$ at all x, which describes a well with no particle. Furthermore, for a negative integer $-n = -1, -2, ..., \psi_{-n}(x) = -\psi_n(x)$, and it differs from $\psi_n(x)$ by an overall sign. But two wavefunctions that different by an overall constant are equivalent with identical probability density when normalized.

The wave vector is discrete, which means that both the momentum and the energy of the particle are discrete too, since $p = \hbar k$ and $k = \sqrt{\frac{2mE}{\hbar^2}}$. Therefore, $\psi_n(x) = A_n \sin k_n x = A_n \sin(\frac{n\pi}{a}x)$. Normalizing and choosing the wavefunctions to be real, $1 = \int_0^a |\psi_n(x)|^2 dx = \int_0^a A_n^2 \sin^2(\frac{n\pi}{a}x)dx = A_n^2 \frac{a}{2}$, or $A_n = \sqrt{\frac{2}{a}}$. The corresponding energies are $E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{\hbar^2\pi^2 n^2}{2ma^2}$. Summarizing, the normalized wavefunctions and the corresponding energies for the particle of mass m in the given

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infinite square well are

$$\psi_n(x) = \sqrt{\frac{2}{a}}\sin(\frac{n\pi}{a}x), \quad E_n = \frac{\hbar^2\pi^2 n^2}{2ma^2}, \qquad n = 1, 2, 3, \ldots \tag{2.2.6}$$

The wavefunctions and the corresponding energies are shown in the schematic figure below.

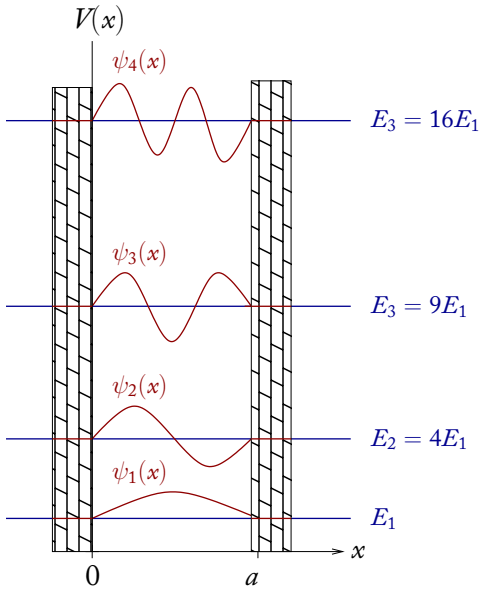


Figure 2.2.2: Wavefunctions and energies for a particle in the infinite square well potential.

Exercise 2.2.1:

1. Demonstrate that the particle cannot have $E = 0$ or $E < 0$ for the given potential well above the $x-$axis.
2. Show that $\psi(x) = 0$, for a particle of finite energy, outside the well by considering a finite potential of height V_0 and taking the limit $V_0 \to \infty$.

Solution:

1. If $E = 0$, then we have $\psi''(x) = 0$. This has the general solution $\psi(x) = Cx + D$. This with the boundary conditions $\psi(0) = \psi(a) = 0$ sets a straight line that passes through both $x = 0$ and $x = a$, or $\psi(x) = 0$, and there is no particle with $E = 0$. For $E < 0$, the Schrödinger equation becomes $\psi''(x) - \kappa^2\psi(x) = 0$, where $\kappa = \sqrt{-\frac{2mE}{\hbar^2}}$. The general solution is $\psi(x) = Ce^{-\kappa x} + De^{\kappa x}$. Applying the boundary conditions $\psi(0) = C + D = 0$ and $\psi(a) = Ce^{-\kappa a} + De^{\kappa a}$. Solving these, we get either $C = D = 0$ or $\kappa a = 0$. Therefore, we cannot have a

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particle with $E < 0$.

2. Considering a finite $V_0 > E$, the general solution is $\psi(x) = Ce^{-\kappa x} + De^{\kappa x}$, where $\kappa = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}$ now. Normalizability requires finiteness of the wavefunction at $x = \pm\infty$. Therefore, $\psi(x) = Ce^{-\kappa x}$ for $x > a$ and $\psi(x) = De^{\kappa x}$ for $x < 0$. Both vanish in the limit $V_0 \to \infty$ or $\kappa \to \infty$.

Exercise 2.2.2:

1. The wavefunctions and energies for a particle of mass m in a one-dimensional infinite square well with potential $V(x) = 0$ within $0 \leq x \leq a$ and $V(x) = \infty$ outside are given by $\psi_n(x) = \sqrt{\frac{2}{a}} \sin(\frac{n\pi}{a}x)$ for $0 \leq x \leq a$, $\psi_n(x) = 0$ outside, and $E_n = \frac{\hbar^2\pi^2n^2}{2ma^2}$, where $n = 1, 2, ...$

   (a) Determine $\langle\hat{x}\rangle$, $\langle\hat{x}^2\rangle$, and Δx in state $\psi_n(x)$. Does Δx increase, decrease, or stay the same with increasing n?

   (b) Determine $\langle\hat{p}\rangle$, $\langle\hat{p}^2\rangle$, and Δp in state $\psi_n(x)$. Does Δp increase, decrease, or stay the same with increasing n?

   (c) Check whether the uncertainty principle holds for $\Delta x \Delta p$ in each state $\psi_n(x)$. Does $\Delta x \Delta p$ increase, decrease, or stay the same with increasing n? (5 points)

   (d) What is ΔE in state $\psi_n(x)$?

   (e) Determine $[\hat{x}, \hat{H}]$. Point out whether $\psi(x) = \hat{x}\psi_n(x)$ is a stationary state or not without explicitly determining $\psi(x, t)$.

   (f) The particle is in state $\psi(x, 0) = \psi_n(x)$ at time $t = 0$.

      i. What is the probability that the particle is found in the left half of the well within $0 \leq x \leq \frac{a}{2}$ at time $t = 0$?

      ii. What is the probability that the particle is found in the left quadrant of the well within $0 \leq x \leq \frac{a}{4}$ at time $t = 0$? Interpret and discuss the difference between even n and odd n.

      iii. What is the state $\psi(x, t)$ and the average energy of the particle at later time t? Do your answers for (1(f)i) and (1(f)ii) change at time t?

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(g) The particle is in state $\psi(x,0) = c_1\psi_1(x) + c_2\psi_2(x)$ at time $t = 0$, where c_1 and c_2 are real constants.

i. Write down the normalized $\psi(x,0)$ and $\psi(x,t)$ at time t.

ii. List all possible values of energy that a single measurement could produce at time t and the corresponding probabilities. What is the expectation value of the energy (or average energy) of the particle at time t?

iii. Show that the probability density for finding the particle at point x within the well at time t oscillates in time. Determine the frequency and the amplitude of oscillation, and point out how the frequency is related to the energies E_1 and E_2.

iv. Determine $\langle\hat{x}\rangle$ at time t.

v. Determine $\langle\hat{p}\rangle$ at time t.

(h) The particle is in state $\psi(x,0) = \sqrt{\frac{30}{a^5}}\,x(a-x)$ for $0 \le x \le a$ and $\psi(x,0) = 0$ outside the well at time $t = 0$.

i. Make a schematic plot of $\psi(x,0)$. Determine the state $\psi(x,t)$ of the particle at later time t.

ii. What are the possible values that a single measurement of energy would produce at time t and the corresponding probabilities. Point out the energy with the highest probability and interpret and discuss the reason or what it means.

(i) Rewrite the wavefunctions when the location of the well where $V(x) = 0$ is moved to within $-\frac{a}{2} \le x \le \frac{a}{2}$.

Question 2.2.2: For the wavefunctions of the infinite square well, (A). $\psi(x)$ is continuous, (B). $\psi'(x)$ is continuous, (C). $\psi''(x)$ is continuous.

Answer: The answer is only (A). $\psi(x)$ is continuous. Because the potential has infinite discontinuity at $x = 0$ and $x = a$, so does $\psi''(x)$ while $\psi'(x)$ has a finite discontinuity. On the other hand, $\psi(x)$ is continuous with only a kink at $x = 0$ and $x = a$. Observe that the wavefunctions have values of zero (and are flat) outside the well and start with nonzero slope at the well boundaries. Alternatively,

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$\psi'(0) = 0$ for $x \leq 0$ and $x \geq 0$ while $\psi'(x) = \sqrt{\frac{2}{a}}\frac{n\pi}{a}\cos(\frac{n\pi}{a}x) \neq 0$ at $x = 0$ and $x = a$ and $\psi'(x)$ discontinuous at $x = a$. Moreover, $\psi''(x)$ is discontinuous because $\psi''(x) = \frac{2mE}{\hbar^2}V(x)\psi(x)$ and $V(x)$ is discontinuous while $\psi(x)$ is continuous at $x = 0$ and $x = a$. However, note that the kink in $\psi(x)$ and the discontinuity in $\psi'(x)$ is because of the artificial nature of the potential that is infinite outside the well, which cannot be realized exactly in the real world. Therefore, in the real world, no potential jumps by an infinite amount, and both $\psi(x)$ and $\psi'(x)$ would be continuous.

Question 2.2.3: For a single particle in a symmetric potential, the ground state wavefunction is (A). symmetric. (B). antisymmetric. (C). either symmetric or antisymmetric.

Answer: The answer is (A). symmetric. We know that the stationary wavefunctions can be chosen either symmetric or antisymmetric. In fact, for a particle inside a symmetric potential well, the stationary states that satisfy the boundary conditions are automatically either symmetric or antisymmetric. Now the ground state is the state of lowest energy (and lowest frequency). The wavefunction with lowest energy oscillates least and can satisfy the boundary conditions without a node in the interior of the potential. Therefore, it is symmetric. In order to be antisymmetric, a wavefunction has to cross the $x-$axis and have a node at the middle, and that oscillates more than the ground state wavefunction.

Question 2.2.4: How many nodes does $\psi_n(x)$ of the infinite square well have in the interior $0 < x < a$? (A). 0. (B). 2. (C). n. (D). $n-1$. (E). $n+1$. (F). It cannot be determined.

Answer: The answer is (D). $n-1$. The ground state corresponds to $n = 1$ and has no node in the interior. The first excited state has one node. The nth state has $n-1$ nodes. The point here is that as the energy of a stationary state increases, the wavefunction oscillates and crosses the $x-$axis more.

Question 2.2.5: Given stationary states $\psi_1(x,t)$ and $\psi_2(x,t)$, is $\psi_1(x,t) + \psi_2(x,t)$ stationary? (A). Yes. (B). No.

Answer: The answer is (B). No. A state is stationary if the probability density is time-independent. Now $\psi_1(x,t) + \psi_2(x,t) = \psi_1(x)e^{-\frac{i}{\hbar}E_1t} + \psi_2(x)e^{-\frac{i}{\hbar}E_2t} = e^{-\frac{i}{\hbar}E_1t}\left(\psi_1(x) + \psi_2(x)e^{-\frac{i}{\hbar}(E_2-E_1)t}\right) = e^{-\frac{i}{\hbar}E_1}\left(\psi_1(x) + \psi_2(x)e^{-i\omega t}\right)$, where $\omega = \frac{E_2-E_1}{\hbar}$. There-

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fore, taking real $\psi_1(x)$ and $\psi_2(x)$, $|\psi_1(x,t)+\psi_2(x,t)|^2 = \psi_1(x)^2+\psi_2(x)^2+2\psi_1(x)\psi_2(x)\cos\omega t$, which oscillates in time.

Question 2.2.6: Given $\psi_1(x,0) = x\psi_n(x)$ for the infinite square well, (A). $\psi(x,t) = x\psi_n(x)e^{-i\frac{E_n}{\hbar}t}$. (B). $\psi(x,t) = x\psi_n(x)e^{i\frac{E_n}{\hbar}t}$. (C). $\psi(x,t) = x\psi_n(x)$. (D). None of the above.

Answer: The answer is (D). None of the above. Because $x\psi_n(x)e^{-i\frac{E_n}{\hbar}t}$ is not one of the stationary states, $\hat{H}\left(x\psi_n(x)e^{-i\frac{E_n}{\hbar}t}\right) \neq E_n(x\psi_n(x)e^{-i\frac{E_n}{\hbar}t})$, we cannot write $\psi_n(x,t1) = \psi_n(x,0)e^{-i\frac{E_n}{\hbar}t}$. [Indeed, it is a linear combination of the stationary states, $\psi(x,0) = x\psi_n(x) = \sum_{m=1}^{\infty} c_m\psi_m(x)$, where $c_m = \int_0^a \psi_m(x)x\psi_n(x)dx = \frac{4amn\left((-1)^{m+n}-1\right)}{\pi^2\left(m^2-n^2\right)^2}$ for $m \neq n$ and $c_n = \frac{a}{2}$. Therefore the probability density evolves in time, or $\psi(x,t) = \frac{a}{2}\psi_n(x)e^{-\frac{i}{\hbar}E_nt} + \sum_{m\neq n}\frac{4amn\left((-1)^{m+n}-1\right)}{\pi^2\left(m^2-n^2\right)^2}\psi_m(x)e^{-\frac{i}{\hbar}E_mt}$ and the state is not stationary.]

Question 2.2.7: Assuming that a quantum dot can be approximately described by an electron in a one-dimensional infinite square well potential, which solution of quantum dots has a well of largest width?



Figure 2.2.3: Quantum dots

(A). Red. (B). Orange. (C). Green. (D). Blue. (E). Violet.

Answer: The answer is (A). Red. The energy levels have of a particle of mass m in an infinite square well of width a are given by $E_n = \frac{\hbar^2\pi^2n^2}{2ma^2}$, which increase with decreasing a. Assuming that light is emitted when an electron makes a quantum transition from a higher energy level of quantum number n to a lower energy level of quantum number m, the energy of the photon emitted is $E = h\nu = h\frac{c}{\lambda} = E_n - E_m = \frac{\hbar^2\pi^2}{2ma^2}(n^2-m^2)$. Considering different widths, we have $h\frac{c}{\lambda} = 0.38eV(n^2-m^2)$ for $a = 1nm$, $h\frac{c}{\lambda} = 0.09eV(n^2-m^2)$ for $a = 2nm$, and $h\frac{c}{\lambda} = 0.01eV(n^2-m^2)$ for $a = 6nm$. As an example if we use the $a = 6nm$ quantum dot to produce a red

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light of wavelength $\lambda = 740nm$, which has a photon energy of $h\frac{c}{\lambda} = 1.68eV$, we need $n^2 - m^2 = \frac{1.68eV}{0.01eV} = 168$. This can be achieved by a transition from $n = 13$ to $m = 1$. In order to be able to implement this, we need to make energy levels between $m = 1$ to $n = 13$ forbidden while allowing energies in a region starting from $n = 13$ allowed (called conduction band). Such energy gaps and bands are produced by making use of appropriate semiconductor materials. One material that is used to produce quantum dots is CdSe/ZnS, a Cadmium Selenide core with Zinc sulfide shell inside a protective oleic acid colloidal solution. The material is three-dimensional. However, the energy levels have similar structure as the infinite square well. The electrons are normally in the ground state at $n = 1$. One starts with shining ultraviolet light that lifts them from the ground state to a higher energy level above $n = 13$ in our example. The electrons then lose some of their energy and fall to the $n = 13$ level. They then fall from the $n = 13$ level to $m = 1$ (since the levels in-between have been disallowed by the gap produced) emitting a photon of the desired red color of monochromatic light.

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